



TOPIC: MINI PROJECT SYNOPSIS ON RENTEASY — VEHICLE/EQUIPMENT RENTAL PLATFORM (MERN STACK)

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DECLARATION

We hereby declare that the project synopsis entitled "**RentEasy — Vehicle/Equipment Rental Platform**" submitted to the Department of Computer Science & Engineering of GIA University Mathura is an authentic record of our own work. This project will be designed and developed using the MERN stack (MongoDB, Express.js, React, Node.js) following clean coding standards, REST API design, and version control best practices under guidance.

Team Member: Nitin Kumar

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INTRODUCTION

About the Project

RentEasy is a full-stack web-based rental marketplace designed to connect equipment/vehicle owners with renters. The platform enables individual owners and commercial vendors to list their vehicles or heavy/light equipment for rent. Customers can search, filter, check availability for specific date ranges, and book equipment seamlessly. Built on the MERN stack (MongoDB, Express.js, React.js, Node.js), RentEasy provides real-time rental management, secure data handling, and transparent booking procedures.

PROJECT REASON & OBJECTIVES

Primary Reason to Choose This Project

- **Market Need:** Traditional vehicle and equipment rental processes suffer from fragmented communications, lack of transparent date-range availability, and inefficient inventory management for small-scale owners.
- **Skill Acquisition:** Developing a end-to-end MERN stack platform provides real-world engineering experience in asynchronous operations, complex database relationships, REST API design, JWT security, and Git version control practices.

The Main Objective of the Project

- Build a responsive web application allowing users to rent vehicles/equipment for custom date ranges.
- Create a dedicated management dashboard for equipment owners to list items, adjust pricing, and track availability.
- Implement robust backend APIs with full CRUD functionality, authentication, and structured validation.
- Maintain clean coding guidelines and Git/GitHub collaborative workflows.

SCOPE & METHODOLOGY

Scope of the Project

- **User Roles:** Renters (customers), Equipment Owners (vendors), and Platform Administrators.
- **Core Functions:** Date-range availability checks, inventory management, user authentication, reservation history, and dynamic searching/filtering.
- **Future Extensions:** Payment gateway integration (Stripe/Razorpay), real-time chat, and location-based GPS tracking.

Working Methodology of the Project

- **Agile Software Development:** The project execution follows iterative sprints covering requirement gathering, database design, API engineering, UI design, component integration, and testing.
- **Version Control:** Git feature-branching workflow with GitHub pull requests to ensure clean code review and integration.

SYSTEM, HARDWARE & SOFTWARE REQUIREMENTS

System Requirements

- **Supported Operating Systems:** Windows 10/11, macOS 11+, or Linux (Ubuntu 20.04+)
- **Software Required:**
 - **IDE/Code Editor:** Visual Studio Code
 - **Runtime Environment:** Node.js (v18.x or higher)
 - **Database:** MongoDB (Local instance or MongoDB Atlas)
 - **API Testing Tool:** Postman
 - **Version Control:** Git & GitHub

Hardware Requirements

- **Processor:** Intel Core i3/i5 or AMD Ryzen 3/5 (Quad-Core minimum)
- **RAM:** 8 GB minimum (16 GB recommended)
- **Storage:** 20 GB free disk space (SSD recommended)
- **Display:** 1366x768 resolution or higher

Testing Technology

- **Backend API Testing:** Postman, Jest, Supertest
- **Frontend UI Testing:** React Testing Library, Chrome DevTools

FRONTEND AND BACKEND STACK

Frontend

- **React.js:** Single-page application framework for dynamic user interfaces.
- **Axios / Fetch API:** Asynchronous HTTP requests to the Node.js backend.
- **Tailwind CSS / Bootstrap:** Responsive styling and mobile-friendly layouts.

Backend

- **Node.js & Express.js:** Event-driven RESTful API server.
- **MongoDB & Mongoose:** NoSQL database schema modeling for flexible document storage (users, listings, bookings).
- **JWT (JSON Web Tokens) & Bcrypt:** Secure user authentication and password hashing.

MODULE DESCRIPTION

1. User Management & Authentication Module

- Registration and login for Renters and Owners.
- Role-based access control (RBAC) via JWT middleware.

2. Listing & Inventory Module (Owner)

- Add, update, delete, or hide vehicle/equipment listings.
- Upload item images, set daily rental rates, specify categories, and mark vehicle availability.

3. Search & Booking Module (Renter)

- Filter listings by category, location, and price range.
- Date-range availability picker preventing double bookings.
- Booking confirmation and history management.

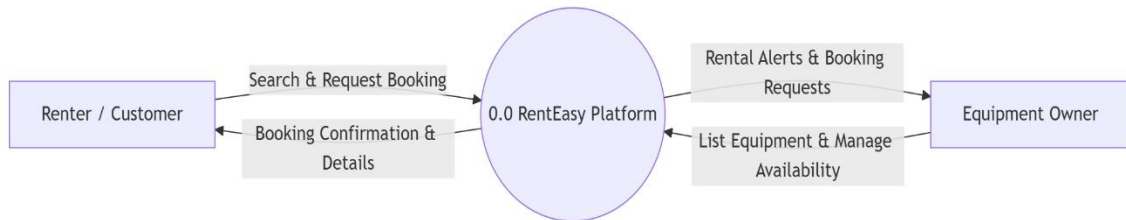
4. Admin Dashboard Module

- Platform monitoring, listing verification, user management, and conflict handling.

DATA FLOW DIAGRAMS (DFD)

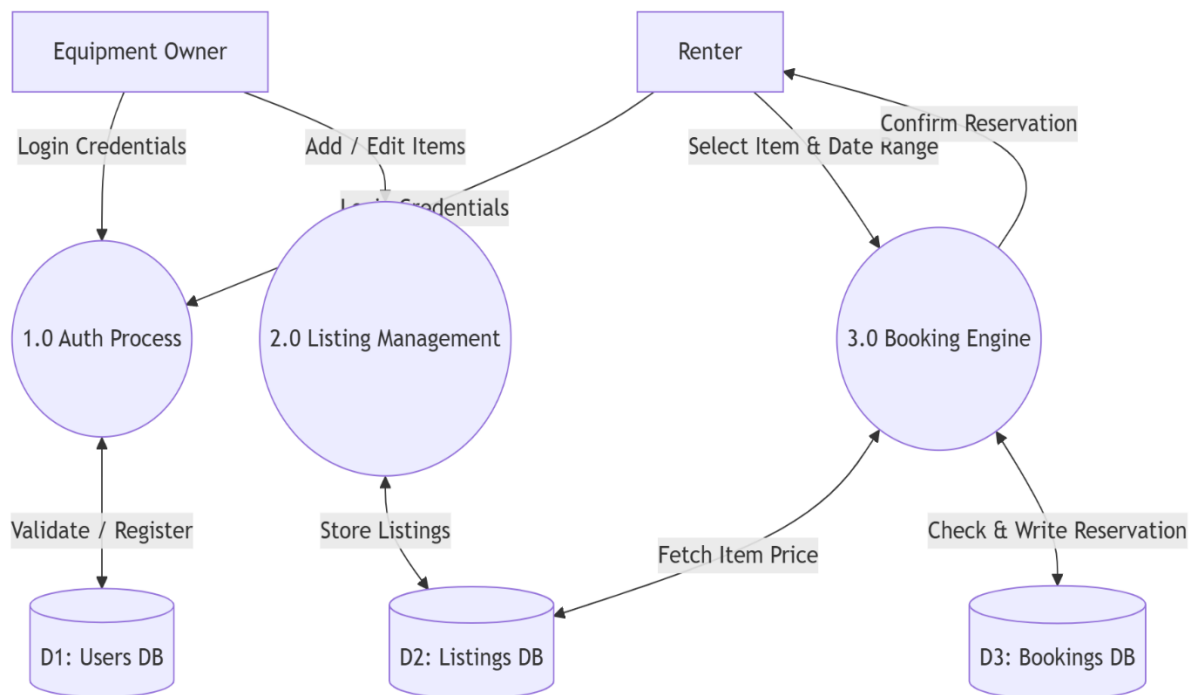
0-Level DFD (Context Level)

The overall system boundary where Users (Renters/Owners) interact with the RentEasy Platform.



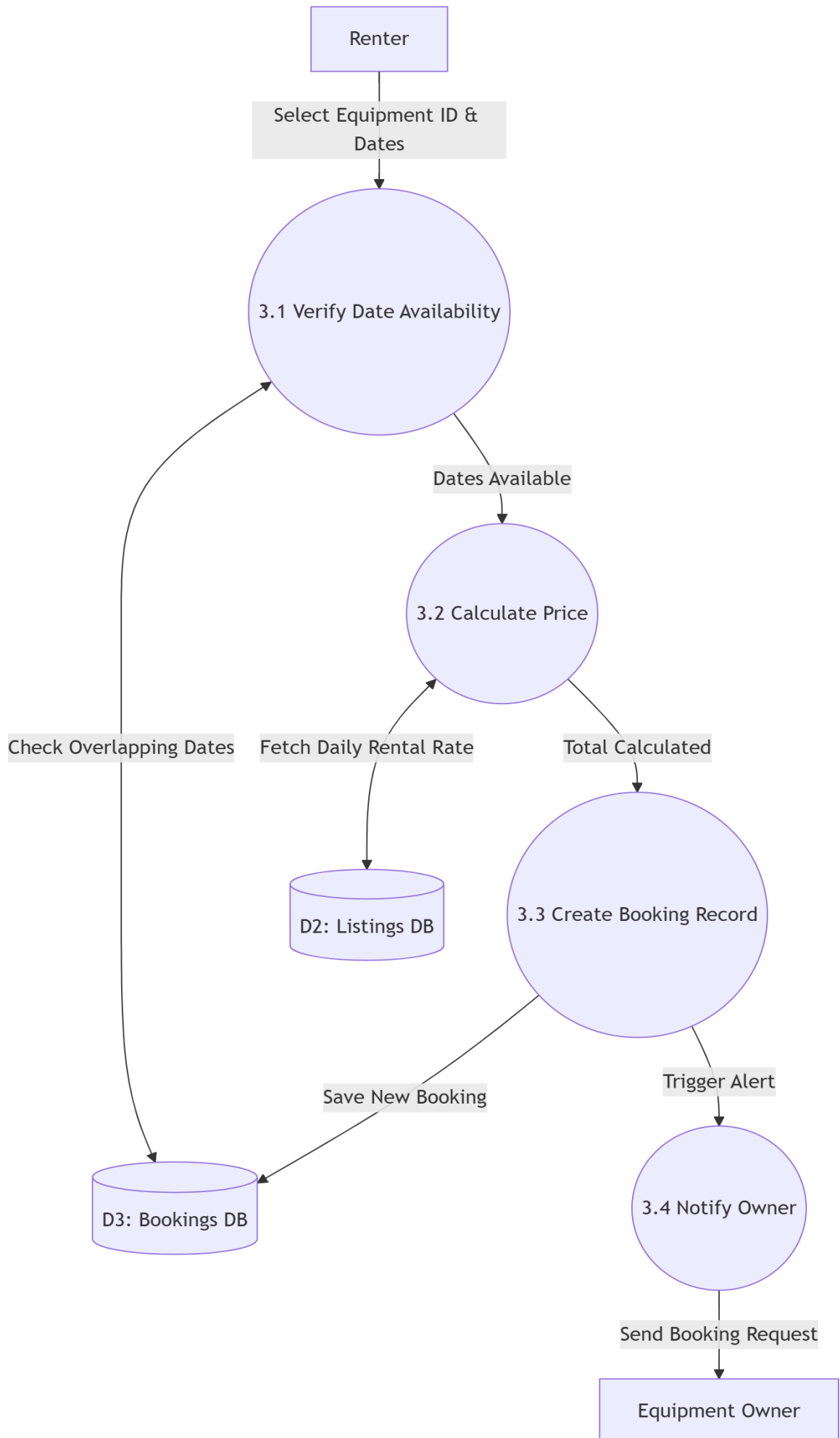
1-Level DFD

Deconstructs the main processes: User Authentication (1.0), Item Management (2.0), Booking & Date Validation (3.0), and Database Storage (4.0).



2-Level DFD (Booking Sub-system)

DATA FLOW DIAGRAMS (DFD)



REFERENCES & BIBLIOGRAPHY

❓ MongoDB Documentation:

<https://www.mongodb.com/docs/>

❓ Express.js API Reference: <https://expressjs.com/>

❓ React Official Documentation: <https://react.dev/>

❓ Node.js Documentation: <https://nodejs.org/en/docs/>

❓ Git Version Control Best Practices: <https://git-scm.com/doc>